

Variables, Data Types, Loops, and Functions in Python

1. Variables

Variables are used to store data values.

```
name = "Praveen"  
age = 25  
salary = 50000.50
```

```
print(name)  
print(age)  
print(salary)
```

Rules for Variable Names

- * Must start with a letter or `\_`
- * Cannot start with a number
- * Case-sensitive (`age` and `Age` are different)

user_name = "John"	Valid
_age = 25	Valid

```
1name = "John"      Invalid
```

2. Data Types

Integer (int)

```
age = 25  
print(type(age))
```

Float

```
price = 99.99  
print(type(price))
```

String (str)

```
name = "Python"
```

```
print(type(name))
```

Boolean (bool)

```
is_active = True  
print(type(is_active))
```

List

```
languages = ["Python", "Java", "JavaScript"]  
print(languages)
```

Tuple

```
colors = ("Red", "Blue", "Green")  
print(colors)
```

Dictionary

```
student = {  
    "id": 1,  
    "name": "Praveen"  
}  
print(student)
```

Set

```
numbers = {1, 2, 3, 4}  
print(numbers)
```

3. Loops

for Loop

Used when the number of iterations is known.

```
for i in range(1, 6):  
    print(i)
```

Output:

```
1  
2  
3  
4  
5
```

Loop Through List

```
fruits = ["Apple", "Banana", "Orange"]
```

```
for fruit in fruits:  
    print(fruit)
```

while Loop

Used when the number of iterations is unknown.

```
count = 1
```

```
while count <= 5:  
    print(count)  
    count += 1
```

break

Stops the loop.

```
for i in range(1, 10):  
    if i == 5:  
        break  
    print(i)
```

continue

Skips the current iteration.

```
for i in range(1, 6):  
    if i == 3:  
        continue  
    print(i)
```

4. Functions

Functions are reusable blocks of code.

Simple Function

```
def greet():  
    print("Welcome to Python")
```

```
greet()
```

Function with Parameters

```
def greet(name):  
    print("Hello", name)
```

```
greet("Praveen")
```

Function with Return Value

```
def add(a, b):  
    return a + b
```

```
result = add(10, 20)  
print(result)
```

Default Parameters

```
def greet(name="Guest"):  
    print("Hello", name)
```

```
greet()  
greet("Praveen")
```

Keyword Arguments

```
def student(name, age):  
    print(name, age)  
  
student(age=25, name="Praveen")
```

Lambda Function

```
square = lambda x: x * x  
  
print(square(5))
```

Quick Interview Questions

****Q1:**** Difference between `for` and `while` loop?

****Answer:**** `for` is used when iterations are known; `while` is used when iterations depend on a condition.

****Q2:**** What is a function?

****Answer:**** A reusable block of code that performs a specific task.

****Q3:**** What is the difference between `return` and `print`?

****Answer:**** `return` sends a value back to the caller; `print` only displays output.

****Q4:**** What is a lambda function?

****Answer:**** An anonymous one-line function created using the `lambda` keyword.

****Q5:**** Is Python statically typed or dynamically typed?

****Answer:**** Python is dynamically typed; variable types are determined at runtime.